BME54701 · AI-BASED NEURAL DATA SCIENCE · 2026-2

Iowa Parkinson's EEG Cohort

One cohort, four sessions, 14 papers

PART 1 Participants & tasks **PART 2** Related work **PART 3** Data structure **QnA**

One cohort, four tasks

#	TASK	SUBJECTS	LENGTH	TRIALS	MEASURES
1	Rest	149	146 s	—	Spontaneous activity, eyes open
2	Simon	147	880 s	384	Response conflict
3	Oddball	146	741 s	240	Novelty detection
4	Interval timing	139	1413 s	80	Time production

- Every recording ran through the same pipeline
- **134 people completed all four tasks** (91 PD, 43 control)
- The same people across all four entries, recorded in one visit
- 64-ch Brain Vision · 500 Hz · Pz reference · all PD on medication

Participants

*MoCA: Montreal Cognitive Assessment
* Unified Parkinson's Disease Rating Scale, Part III

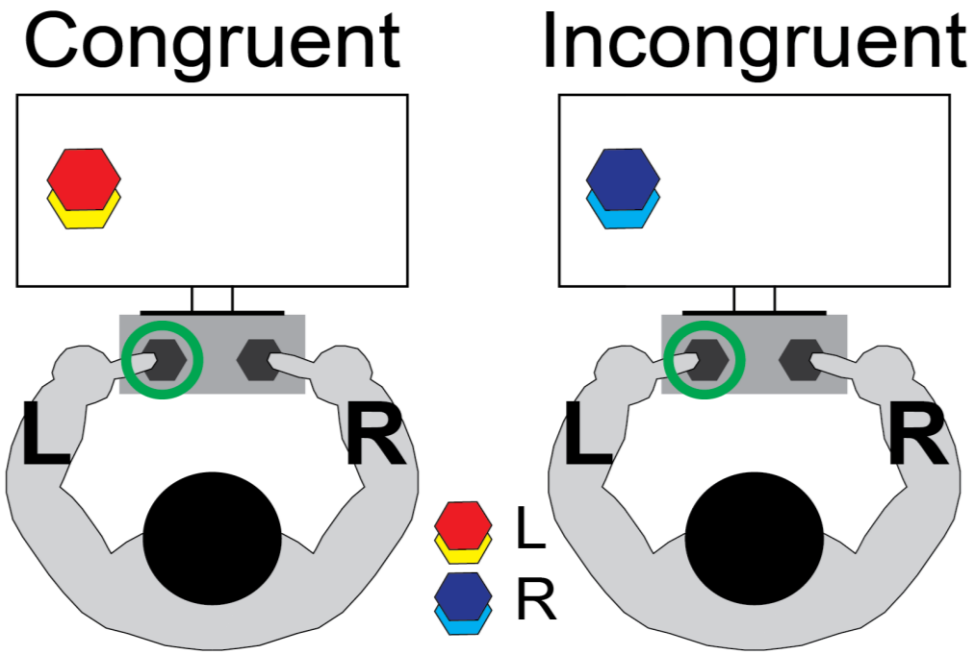
	PD	CONTROL
n	100	49
Age	68.5 ± 8.0	70.9 ± 7.5
MoCA (out of 30)	24.3 ± 4.0, range 9–30	26.7 ± 1.8
UPDRS-III	12.6 ± 7.1	0 by definition

Task 1 — Rest: two minutes, no events

- Eyes open, sitting quietly, about two minutes — median 146 s (range 121–343 s)
- No task and no event markers — only continuous EEG

Task 2 — Simon: respond by colour, not position

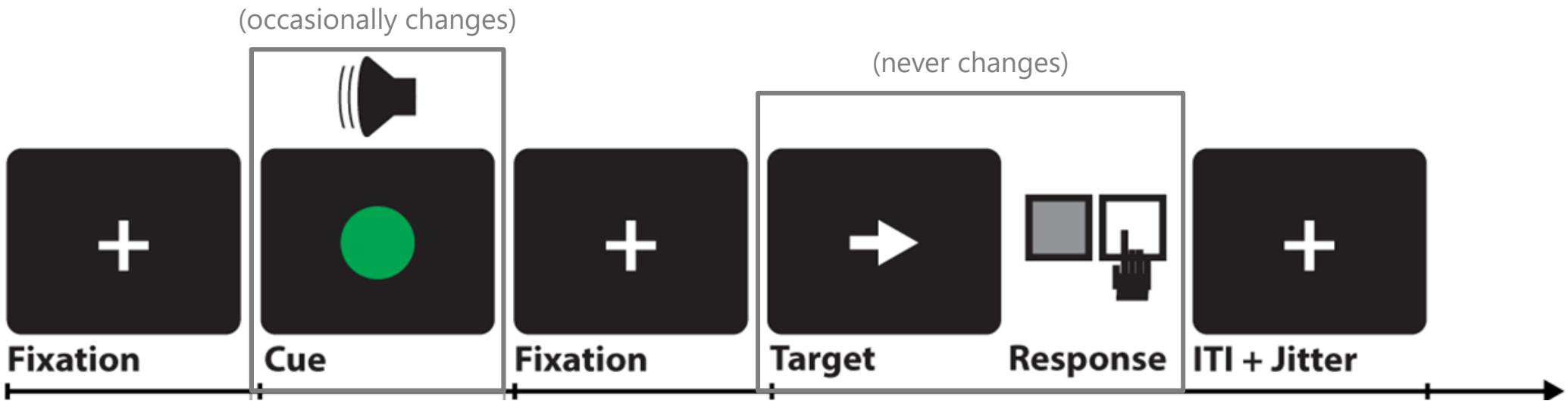
- A colored square appears on the **left or right** of the screen
- Respond by colour: yellow or red → left key, cyan or blue → right key
- **Congruent**: side matches the correct hand | **Incongruent**: opposite side
- 192 trials per condition, 384 total



Sun X et al. arXiv:2410.06534 (2024) Fig 1a · CC BY 4.0

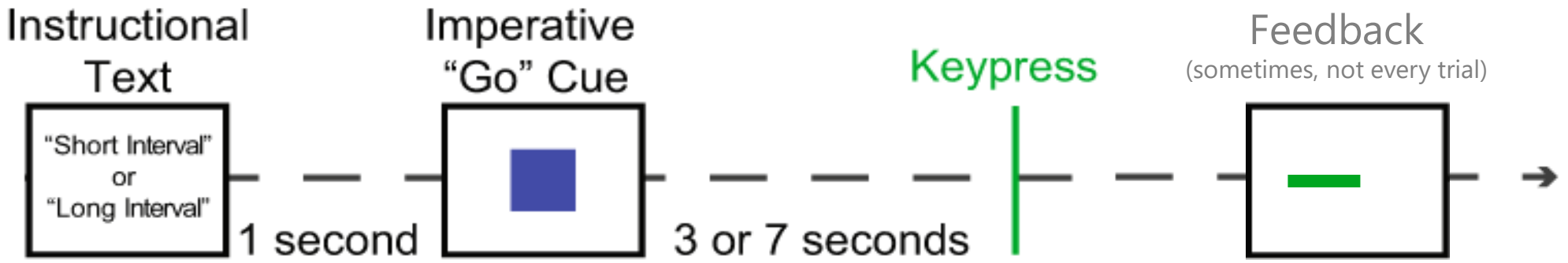
Task 3 — Oddball: the surprise is in the pre-cue

- A **pre-cue** is presented in two modalities at once (visual + sound)
- 464 ms later a **white arrow** appears; press the key it points to
- On a minority of trials the visual or auditory pre-cue is **unexpected**
  - The arrow task never changes → novelty is task-irrelevant
- 240 trials: 192 standard, 24 visual oddball, 24 auditory oddball



Task 4 — Interval timing: produce 3 s or 7 s

- Text for 1 s: **"Short interval"** (3 s) or **"Long interval"** (7 s)
  - The actual durations were **never told** to participants — learned from feedback
- A box appears; press the spacebar when the interval feels elapsed
- Instructed not to count; distractor vowels appear at random times
- 80 trials, 40 per condition



Singh A et al. npj Parkinsons Dis 7:14 (2021) Fig 1a · CC BY 4.0

Three tracks, one claim

PAPERS	WHAT THEY DO
3	**Task specs and first results.** By the people who made the data. Trial counts, stimulus rules and analysis windows exist only here.
4	**Pushed to clinical endpoints.** Outcome is MoCA, UPDRS or survival — not accuracy.
7	**Consumed as an ML benchmark.** No new neurophysiology. Model comparison only.

All three original papers make one claim. Parkinson's patients fail to raise midfrontal 1-8 Hz to a cue, in proportion to cognitive decline. ρ 0.34 with MoCA, ρ 0.46 with executive function.

*MoCA: Montreal Cognitive Assessment

One paper per task

TASK	PAPER	FINDING	NUMBER
Rest	**Anjum 2024** npj PD · original	LPC + PCA + distance ratio. **Not deep learning, still ahead of it.** 8 electrodes, 2.5 min. **Only external validation in the set**.	**AUC 0.90** external ρ 0.68
Simon	**Sun 2024** EMBC · external	Connectivity **between** electrodes beats single-channel waveforms and reaction time. Same features cannot separate congruent from incongruent — **51%**.	**95%** trial-level split
Oddball	**Avola 2025** DMSVIVA · external	11 models, one pipeline, one split. **XGBoost beat every deep model.** No foundation model was actually run.	**0.79** AUC 0.86
Timing	**Singh 2021** npj PD · original	Timing variability links signal to cognition. The only task where **behaviour alone gives a baseline**.	**ρ -0.39** MoCA · d = 0.66

*MoCA: Montreal Cognitive Assessment

Three things to know first

• **Raw OpenNeuro has no condition labels.** Simon and oddball markers are cue and response only — congruent/incongruent and standard/deviant are absent from events.tsv. **The course distribution adds them in trial_info/.**

• **11 of 49 controls score MoCA below 26.** PD-vs-control and impaired-vs-normal are different problems. Do not compare accuracies across the two.

• **Use UPDRS inside the PD group only.** Controls are stored as 0, so a PD-versus-control comparison on UPDRS is meaningless.

> **"Analyze the Simon conflict effect" is not possible with this data alone.**

What is in the distribution

```
IowaPD_clean/
  Dataset_Guide.pdf  this document
  subjects.csv       one row per subject: group, MoCA, UPDRS, ID mapping

  continuous_data/   581 recordings, the canonical data      (.set + .fdt)
  trial_info/        432 condition-label tables              (.tsv)
  epoched_data/      1,149 ready-made epoch files            (.mat)
```

Signal specification

- **64 channels**, identical order in every file, Pz is channel 64
- **500 Hz**, microvolts, **average reference**
- 0.1 Hz high-pass, line noise removed, **no low-pass**

> Rest has no trial_info/ or epoched_data/ because it has no events.

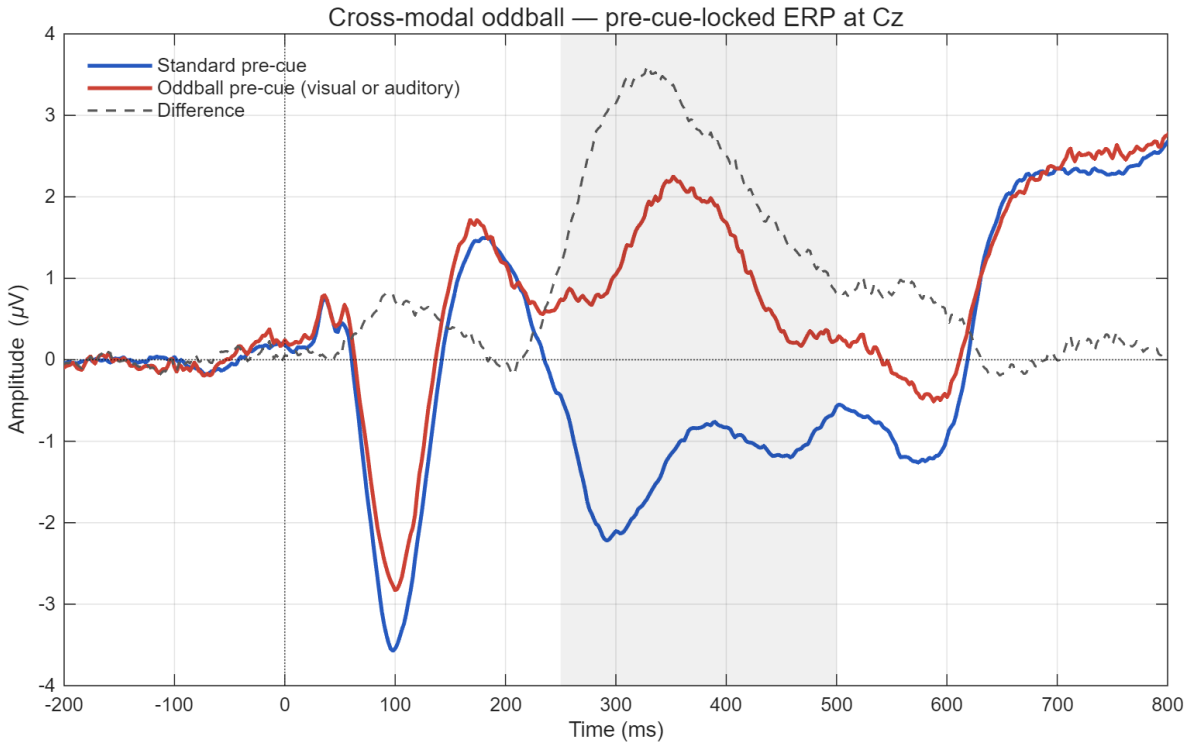
What we did to the data

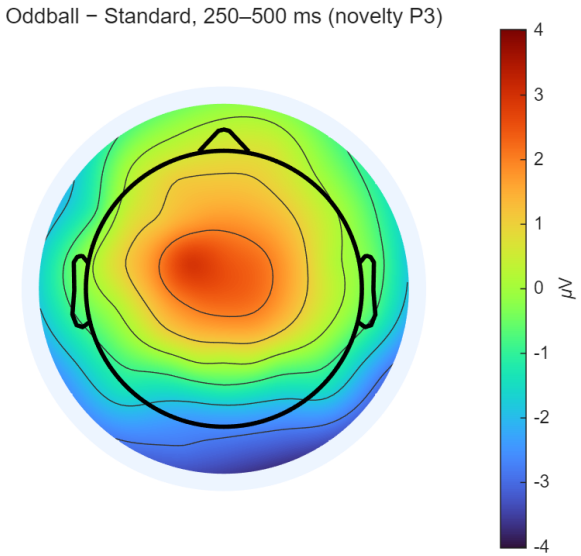
The raw files had no preprocessing at all — only a hardware high-pass

- **Channels** — removed non-EEG channels, repaired mislabeled ones in 39 recordings, replaced electrode coordinates with a standard template
- **Filtering** — line noise removed at 60 / 120 / 180 Hz, 0.1 Hz high-pass
- **Bad channels** — detected with FASTER, then spherical interpolation
  - 1.8 channels per recording on average; Cz never interpolated
- **Reference** — restored Pz, then re-referenced to the 64-channel average
- **ICA** — extended infomax on continuous data, noisy segments excluded from training only
  - eye components removed using ICLabel; 1.6 components per recording on average
- **Epochs** — cut at several anchors; bad epochs **marked but not removed**

> **581 of 581 recordings completed with no failures.** Each recording carries its own processing record in EEG.etc.iowa: which channels were interpolated, which components were removed, and any warnings.

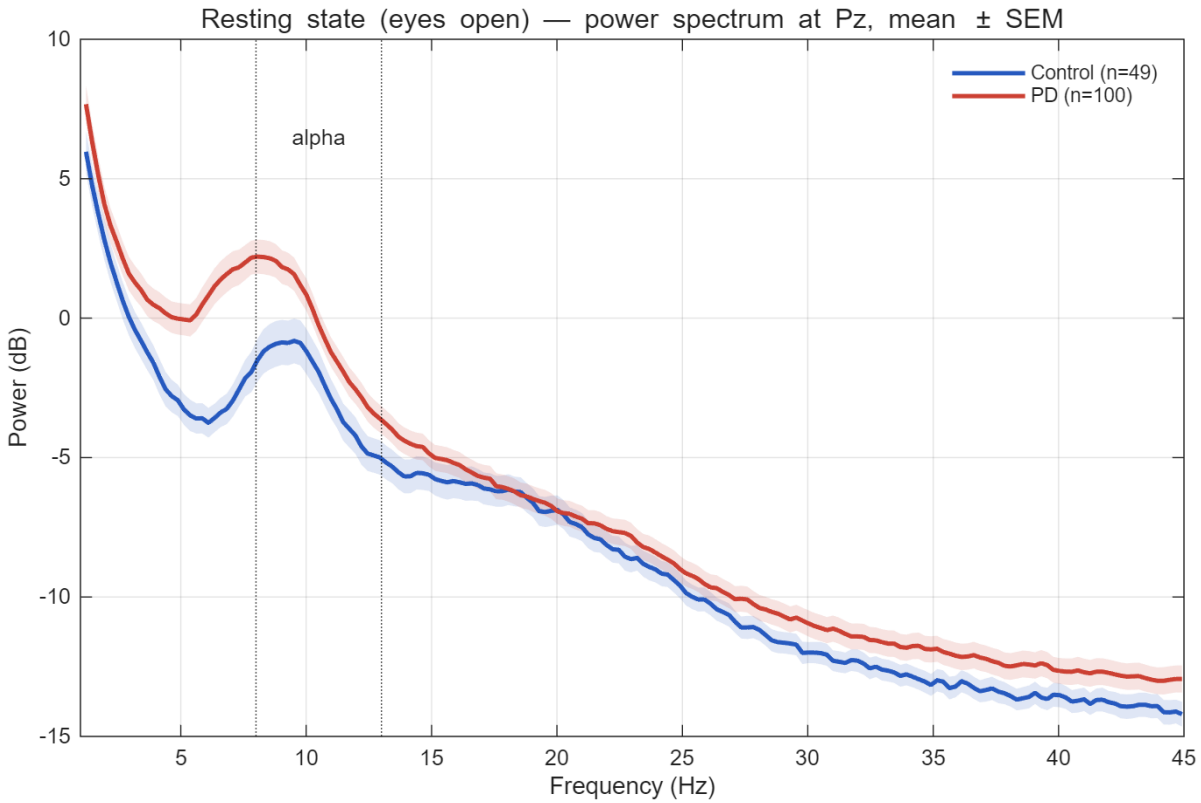
The preprocessing works





Oddball, pre-cue-locked ERP at Cz, grand average over 146 subjects. The oddball minus standard difference peaks near 330 ms with a fronto-central maximum — the standard novelty P3 pattern.

The preprocessing works



Resting state, power spectrum at Pz, mean and SEM. A clear alpha peak near 9–10 Hz in both groups.

- Both figures are in the guide, together with the Simon ERP and time-frequency plots
- Use them as reference: if your own grand average looks nothing like this, check your code first

Where projects can go

Starting points

- Does the Simon conflict effect differ between groups, in RT and in errors?
  - Behavioral tables only, no EEG needed — a good first step
- Is the novelty P3 smaller in PD?
- Does resting alpha power or peak frequency relate to MoCA?

Going further

- Is midfrontal theta during conflict related to cognitive status?
- Does timing variability predict MoCA better than EEG does?
- Visual against auditory novelty: the same response or not?
- Combine measures from all four tasks into one index
- Does a model trained on one task generalize to another in the same person?
- How few channels are enough?
- Resting-state network organization against clinical severity

> **If you build a classifier, split by subject.** If a person contributes to training, none of their data may appear in testing. Splitting by trial or epoch inflates accuracy because the model can recognize individuals rather than the condition.

Where to find things

QUESTION	WHERE
What was each task, what signal to expect	Guide, section 3
Project ideas with difficulty levels	Guide, section 4
What does this column mean	Guide, section 6
What exactly was done to the data	Guide, section 7
Which subjects should I trust	Guide, section 8 and EEG.etc.iowa
How do I do X	Guide, section 9
Is my result reasonable	Guide, section 10

First steps

- Read sections 1 and 2 of the guide, then the task you plan to use
- Run the starter code in section 2.2 of the guide (MATLAB or Python)
- Reproduce one number from section 10 before starting your own analysis

> Cite the original datasets: OpenNeuro ds004574, ds004579, ds004580, ds004584 (University of Iowa, CC0).

Team organization

- **Team**
  - Recommendation: maximum 4 students/team
  - Team size doesn’t affect evaluation
  - Separate teams between graduate and undergraduate
- **Blackboard**
  - An announcement on team formation will be posted soon
  - One representative per team leaves a comment on the post
  - **Deadline: Friday, 11:59 PM**
  - Not yet on a team? Email the TA

Teaching Assistant

- Seungbeom Seo (BME, MS program) — ssb3304+2026BME54701@unist.ac.kr
- Chaeeun Yoon (BME, MS-PhD Combined) — cetgth@unist.ac.kr

QnA